Oral Cancer Screening Report

AI-Assisted Mucosal Tissue Classification | Clinical Research Prototype

23 October 2026 20:29

WHO 5th Ed. Criteria | Swin-T Model

CLASSIFICATION ASSESSMENT

CANCER

Model Confidence: 98.3%

Logit Separation: 6.11

Tissue Focus: Intraoral Mucosa

Report ID: #10232029

Analysed photograph

1. Image Quality & Illumination Assessment

Quality Parameter	Measured Value	Target Standard	Status
Illumination (Luminance)	141 / 255	55 – 225	✓ Pass
Contrast Spread (Std Dev)	48	≥ 16	✓ Pass
Sharpness / Focus (Laplacian)	3535.9	≥ 28	✓ Pass
Class Separation Margin	6.11	≥ 3.2 (In-Distribution)	✓ Pass

2. Diagnostic Explainability (Why This Image Received This Prediction)

1. Mucosal Surface & Lesion: Vision attention detected irregular mucosal discolouration (erythroplakic/ulcerative texture) with localized epithelial lo

2. Margin & Border Distortion: High-frequency patch analysis revealed architectural disruption along tissue boundaries, with loss of smooth anatom

3. Classification Rationale: Elevated neural weighting across malignant mucosal patterns produced a 98.3% classification confidence; requires

3. Clinical Risk Interpretation Guide

AI Output	Clinical Presentation & Pathology	Recommended Clinical Pathway
NORMAL / BENIGN	Unremarkable mucosal tissue; no malignant markers	Continue routine periodic dental examination
CANCER / OSCC	Features consistent with oral mucosal carcinoma	Immediate referral for specialist biopsy & histopathology
INCONCLUSIVE	Out-of-distribution frame or borderline margin	Retake close-up oral photograph under bright direct light

4. Important Medical Notice

NON-DIAGNOSTIC RESEARCH PROTOTYPE: This screening tool is an experimental AI classifier and does NOT constitute a medical diagnosis. Final diagnosis requires clinical evaluation, physical palpation, and histopathology (biopsy) by a qualified oral surgeon or oncologist.